# Technical background to our Moodle sites

# Purpose

## Purpose of these slides

At the heart of our ‘complex information technology solutions’ are web-based systems

We need to understand the core components of these systems

- In these slides the emphasis is on getting an overview
- In the later topics we will look more closely

# Contents overview

## What we are going to look at

The software and hardware involved in principle

More specifics for our Moodle sites

What distinguishes our Moodle sites from a production environment Moodle site (such as Massey's Stream)

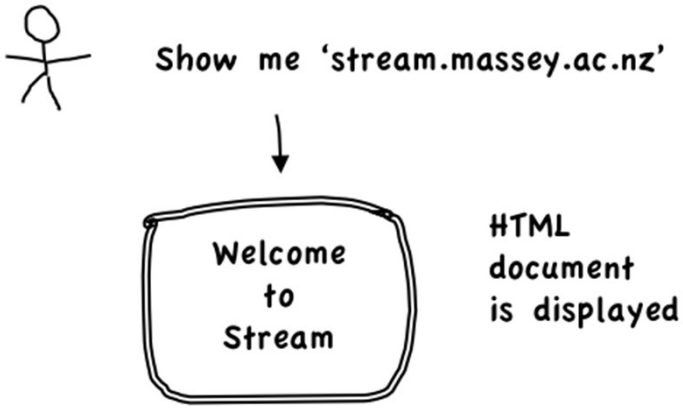
The types of user roles involved

What about putting your own explanations together before reading on?

You will have used many web-based systems – can you explain how they work?

There are many aspects to this and many ways to start explaining – I'll give you one way over the next slides, you might find another way.

# Hardware and software involved

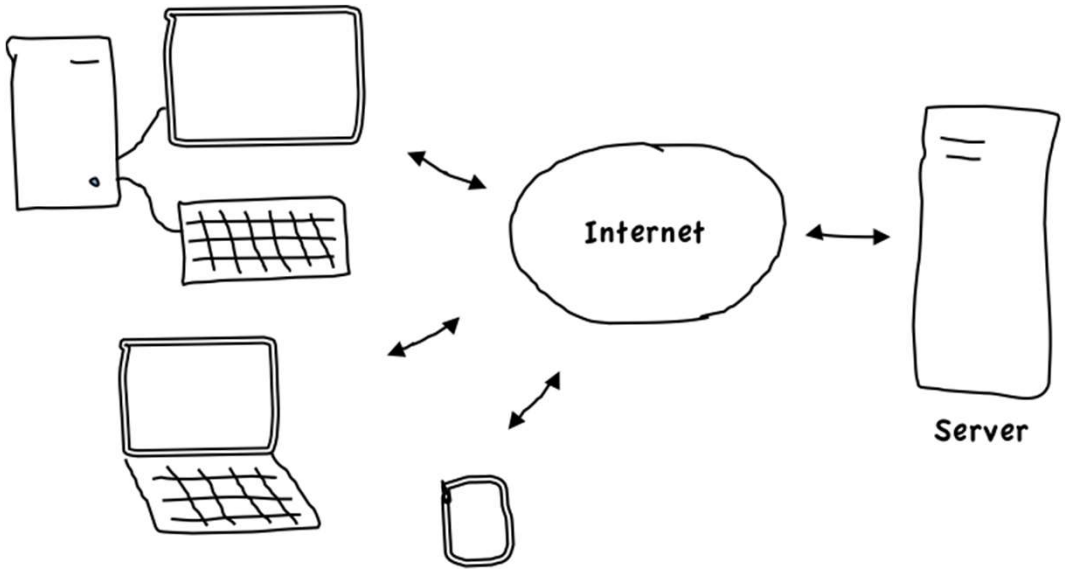



Using a web browser software on our device we request information

This information is displayed to us in response

# Hardware and software involved

Data (our requests and the answers) are transferred via the Internet




We can use a variety of devices at ‘any’ location

Somewhere there is a computer, called a server, that responds to our requests

# Hardware and software involved

Later on we need to look at how this communication via the Internet works

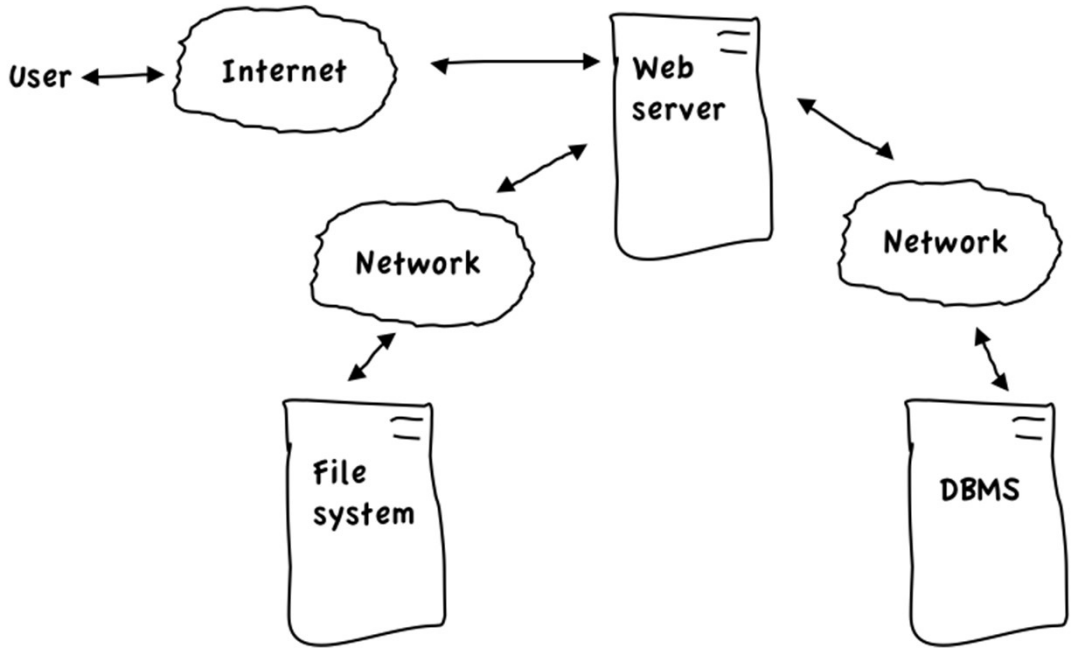



With ‘user’ I am thinking of a person using a web browser on their device

‘System’ is the part we now want to look at more closely

# Hardware and software involved

A web-based system like Moodle has the following key components




The webserver handles the requests coming from the user, collects and formats data to be given back to the user

E.g., the student looks at the discussion postings

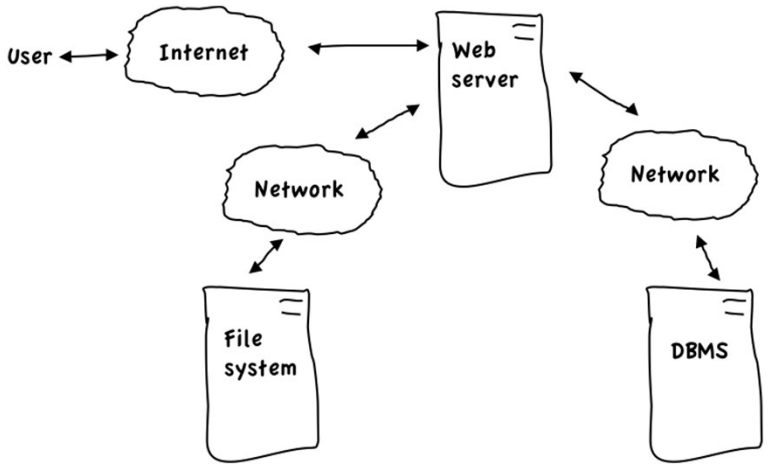
The file system keeps documents

E.g., the lecture slides the teacher makes available, the work the students have submitted as assignments

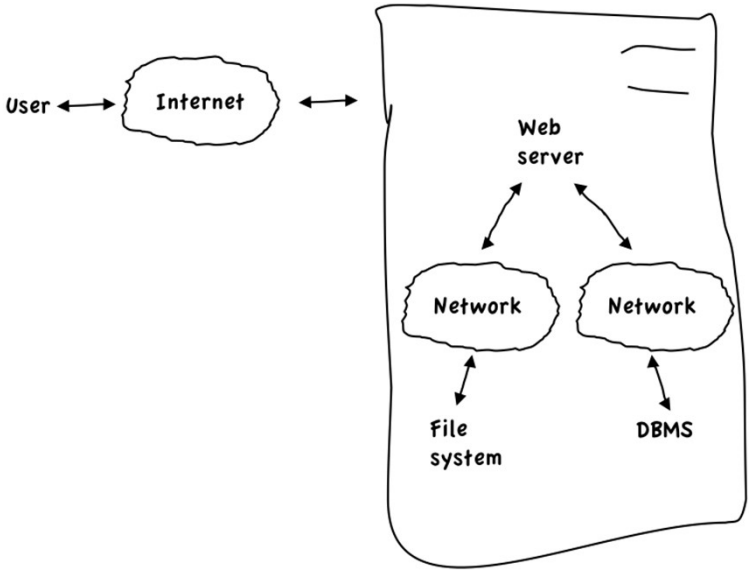
The Data Base Management System stores and manages data

E.g., what courses do we have, what grade did a student get for an assignment

# Hardware and software involved




The webserver software, the DBMS and the file system can ‘sit’ on physically separate computers




Or, these key components can be on the same computer

For our considerations this does not matter, as the components still work together in the same ways.

E.g., we will look at how the webserver of our Moodle system talks to its DBMS

# A comment to language use

## Language used is context dependent (and not always precise)

We might talk about a webserver either as hardware or software

To spell it out can be cumbersome

What is meant should be clear from the context

Someone might say

‘The database returned data for three courses’

This is not precise, as the database is only structured data; the DBMS is the program that does something

Be aware of this, with general material and also the course material (I try to be clear, but will surely slip up)

When discussing the material yourself, be as clear as possible without loosing flow (it gets cumbersome if we always say ‘the webserver program’ instead of just ‘the webserver’)

# Our Moodle sites and where they ‘run’

## A closer look at our Moodle sites

Moodle has the core components of webserver, DBMS and file system

We are using

- An Apache webserver
- A Maria DBMS
- A Linux file system

All of this is running on a server controlled by Massey, accessible to us via the Internet

- We do not see how this server computer is configured or where it is located
- An actual physical box on campus?
- A virtual computer part of the Microsoft Azure cloud environment?

How can several hundred Moodle sites run on the same server?

# Our Moodle sites and where they ‘run’

## How can hundred or more Moodle sites run on the same server?

A webserver has two ‘parts’

The part that is the same regardless of which websites are being served

This includes knowing how to accept a user request and how to send a response back

The part that distinguishes which website is served

This can be a news site, a banking site, a library site

It can be your Moodle site, my Moodle site, Massey’s Stream site

Both parts consist of program files

- Webserver program files (mostly written in C) that have been compiled and are being executed as programs (such as httpd, the HTTP daemon)
- Website specific parts, such as the Moodle program files written in PHP, that get interpreted by the webserver programs (we will look at these PHP files)

# Our Moodle sites and where they ‘run’

How can hundred or more Moodle sites run on the same server?

A DBMS

Has program files that know how to structure and access the data

The data are separated out into different databases

Each of our Moodle sites has its own database

The DBMS knows how to keep these databases separate

We will look at the structures of Moodle databases in general (the part that is common to all Moodle sites)

Each of you will work with your specific Moodle databases (which will contain the data you and your users will create)

# Our Moodle sites and where they ‘run’

## How can several hundred Moodle sites run on the same server?

- The file system
  - Is linked to the operating system of the server
  - Knows how to store files, how to put them into folders, how to find them again
  - It knows how to protect access to these files, letting only authorized users and programs access and modify these files
    - Each of our sites has its own sub-section for its files

The file system keeps the files (like the PHP files) that make up your Moodle site separate from the files that make up my Moodle site

It keeps the user data (like lecture notes or assignments work) on your Moodle site separate from the user data on my Moodle site

Without the file system none of our systems would work

We will use the services of the files system to look at files related to our Moodle sites

# Our Moodle sites and where they ‘run’

## This was one perspective

We have looked at how it is possible to have all our Moodle sites on one computer

You might want to reflect on that (and get it clear in your mind)

## From now we are going to be more self-centred

You focus on

- The files that make up your Moodle site
- The database that supports your Moodle site

This is like pretending that we have the webserver, the DBMS and the whole file system (and the whole server computer) for ourselves

This will not be difficult, as you do not have access to the data of other users

Maybe pause here and reflect?

# Difference between our sites and Stream

What is different between our Moodle sites and Massey's Stream site?

Can you think of an answer before turning to the next slide?

# Difference between our sites and Stream

## What is different between our Moodle sites and Massey's Stream site?

On one level, not much

- Both have the same core components, both work in the same way

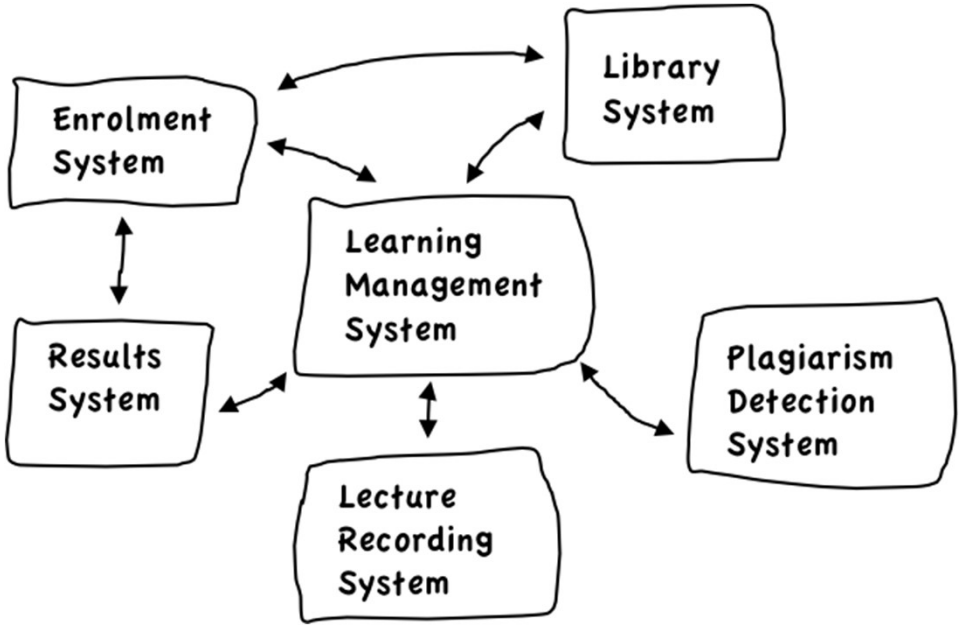
Yet, there are big differences

- Our sites are 'plain vanilla' – no substantial changes or customizations
- Stream is customised
  - Extensions have been installed (e.g., Turnitin for plagiarism checking)
- All of our sites run on one server, Stream runs across several servers
- Safety and security for Stream is very important
- Our sites will not contain any confidential data (please do not enter any)
- Instead of fixing our sites (should we hit any issues) we will simply re-install
- We are admin users for our sites, but have only student access to courses we are enrolled in on Stream (teacher access for me)
- ...

# Looking at system context

## A brief look at the context of a specific Moodle site

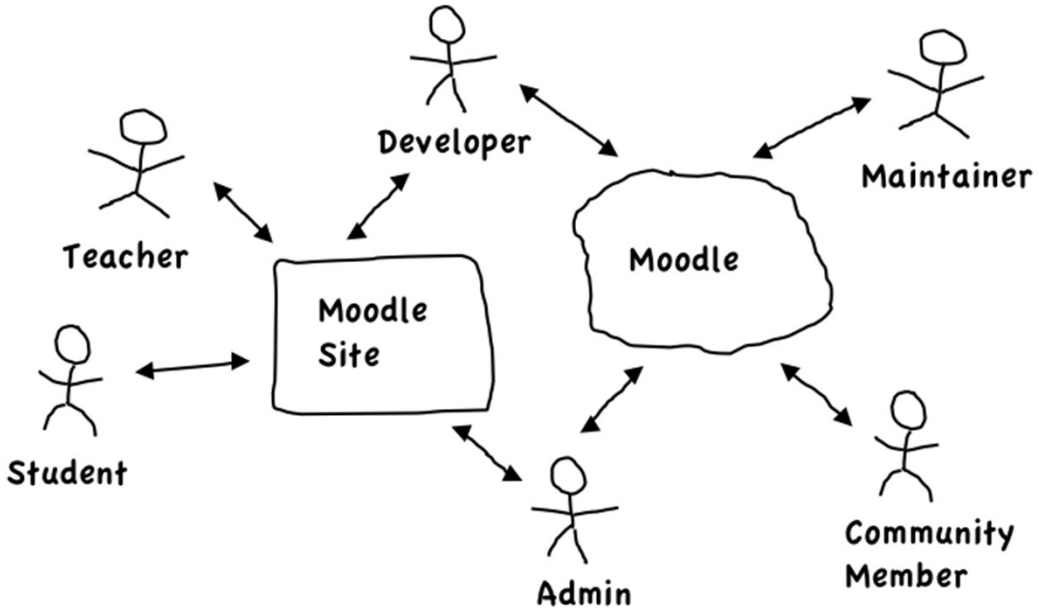
Many systems need to work together to operate an institution like a university



# Looking at the users involved

## Looking beyond a specific Moodle site

Many more types of users/people are involved



# Where to from here

## Write a summary of the material covered in the first week

How you would explain what we have covered to a friend who is not taking this course?

What are the key points, what do they need to know?